Addis Ababa City Administration

Education Bureau

Summer Special Tutorial Program

Condensed Mathematics Curriculum from

Grade 9 – 11

Designed for Grade 11 Students

July 2026 G.C
Addis Ababa/Finfinne

Preface

Dear Teachers,

Addis Ababa City Administration Education Bureau has prepared this Condensed Mathematics Curriculum for students attending the summer special tutorial program (summer camp). The curriculum focuses on core lesson notes, selected content and learning activities aligned with the syllabus. In addition, it includes key examination questions to be covered during this instructional period.

As subject experts, you are encouraged to modify the materials, provide additional explanations and include examples that you believe will help students better understand the concepts, particularly in areas where they commonly experience difficulties. You are also encouraged to draw on your own teaching experience and employ a variety of instructional strategies to keep students actively engaged throughout the program.

Overall, the program emphasizes extensive student practice, active participation, and continuous reinforcement of key concepts. It is designed to ensure that students complete each session with greater confidence and improved competence, thereby enhancing their preparedness for the coming academic year.

How to use this lesson effectively

1. **Begin with the activity**

   Using the provided activity search students' prior knowledge
2. **Explain and elaborate using the key concepts**

   Introduce the topic using the provided summary of the key concepts. These notes are intentionally brief and focused to help students quickly recall core concepts. Feel free to add real-life connections, additional explanations, or alternative methods where necessary.

3. **Use the examples for guided practice**

   Go through the provided examples step by step. Involve students through questioning or board demonstrations to ensure active participation.
4. **Assess understanding before moving on**

   Ask quick oral or written checks—allow students to solve one or two similar questions to confirm their comprehension.
5. **Apply exam-type questions for reinforcement at the end of the week**

   End the session with the exam-format questions at the end of every week. These familiarize students with the **pattern, difficulty level, and expectations** of the national exam.
6. **Adapt as needed**

   Although the structure is uniform, teachers have full flexibility to **add more examples, give extra exercises, or extend discussions** depending on the performance of the class.
7. **Weekly Assessment & Revision (Every Friday)**

Every Friday should be dedicated to **assessment and revision** of the concepts and skills covered during the week.

Suggested Time Allocation for Friday Assessment

- **30 minutes – Assessment:** Students complete the assessment independently under examination conditions.
- **15 minutes – Review and Feedback:** The teacher reviews the answers with the class, explains the correct solutions, addresses common misconceptions and provides constructive feedback.

Lesson 16 (Day 16)

Similarity of plane figures

☞ By the end of this lesson students would be able to:

- State and use the criteria for similarity of triangles
- Solve problems on perimeters and areas of similar triangles and similar polygons

Activity

1. When do we say to figures are similar?
2. Are any two rectangles similar?

Key concepts:

1. Two plane figures are **similar** if their corresponding angles are congruent and the ratios of their corresponding sides are proportional.
2. Similarity theorems for triangles:
   - **AA similarity Theorem:** if the two angles of one triangle are respectively congruent to the two corresponding angles of the other, then the triangles are similar
   - **SSS similarity Theorem**: If the corresponding three sides of two triangles are proportional to each other, then the triangles are similar
   - **SAS similarity theorem**: If the corresponding two sides of two triangles are proportional and the included angles are congruent, then the two triangles are similar

3. If $\Delta ABC \sim \Delta DEF$, then

- $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$ and $m(\measuredangle A) = m(\measuredangle D), m(\measuredangle B) = m(\measuredangle E), m(\measuredangle C) = m(\measuredangle F)$

4. For two similar plane figures:

- $\frac{P_1}{P_2} = \frac{S_1}{S_2}$ and $\frac{A_1}{A_2} = \left(\frac{S_1}{S_2}\right)^2$, where A_1, P_1, S_1 are area, perimeter and side of the first figure and A_2, P_2, S_2 are the corresponding area, perimeter and side of the second figure

Examples

1. Write true if the statement is correct and false otherwise.

a. All squares are similar.

b. All rectangles are similar.

c. Two polygons with the same number of sides are always similar.

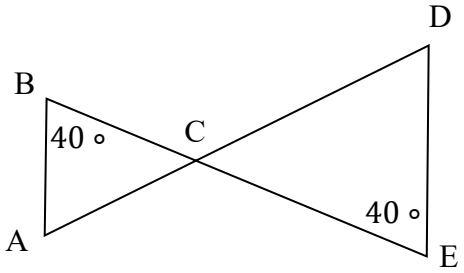
d. If two polygons are similar, their corresponding sides are congruent.

e. If two polygons are similar, their corresponding angles are congruent

Solution:

a. True b. False c. False d. False e. True

2. In Figure below, determine whether the two given triangles are similar



Solution:

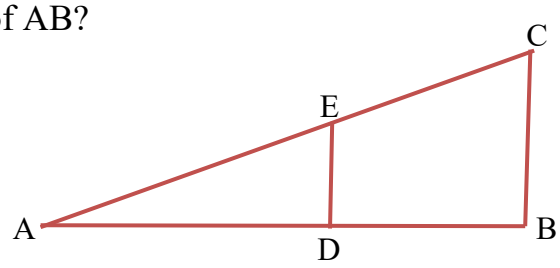
- $m(\measuredangle ABC) \cong m(\measuredangle DEC)$ …….. Given
- $m(\measuredangle ACB) \cong m(\measuredangle DCE)$…….vertical opposite angles. Hence by AA similarity theorem $\Delta ABC \sim \Delta DEC$

Exercise

1. The sides of a polygon have lengths 5, 7, 8, 11, and 19 units. The perimeter of a similar polygon is 75 units. Find the length of the largest side of the larger polygon.

2. The areas of two similar triangles are $144cm^2$ and $81\ cm^2$. If one of the sides of the smaller triangle is $6\ cm$ long, what is the length of the corresponding side of the larger triangle?

3. In the figure below $\overline{DE} \parallel \overline{BC}$. If $AE = 10cm, EC = 5cm\ and\ DB = 4cm$, then what is the length of AB?



Exam type questions

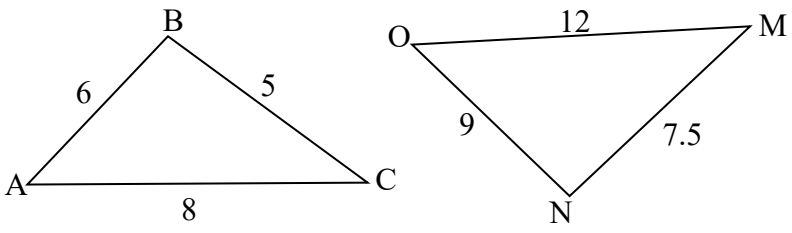
1. Suppose $\Delta ABC \sim \Delta DEF$ and the sides of ΔABC are 14 cm, 7cm, and 19cm long. If the perimeter of ΔDEF is 160cm, then what is the length of the shortest side of ΔDEF?

   A. $56cm$ B.$14cm$ C. $112cm$ D.$28cm$

2. Let $\Delta ABC \sim \Delta DEF$, the areas of ΔABC and ΔDEF are $256m^2$ and $64m^2$ respectively. Which one of the following statement is true?
   A. The ratio of their corresponding sides is 4
   B. If $CA = 9m$, then $FD = 3m$
   C. The ratio of their area is 16
   D. The ratio of their perimeter is 2

3. A student found out that the following two triangles are similar by SSS similarity theorem.

   A. $\Delta ABC \sim \Delta ONM$
   B.$\Delta ABC \sim \Delta MON$
   C.$\Delta ABC \sim \Delta NMO$
   D. $\Delta ABC \sim \Delta OMN$



Lesson 17 (day 17)

Functions and composition of functions

☞ By the end of this lesson students would be able to:

- Define a function
- Find the compositions of two functions

Activity

1. What is function?

2. Is the relation $R = \{(3,1), (2,1), (0,5), (3,1)\}$ a function?

Key concepts:

- A **function** is a relation such that no two different ordered pairs have the same first-coordinates and different second-coordinates.
- If f: A $\rightarrow$ B is a function, then, for any $x \in$ A the image of x under f, $f\ (x)$ is called the functional value of f at x
- Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be functions, then the function h defined by $h(x) = g(f(x))$ for $x \in A$ and $f(x)$ is in the domain of g is called the composition of g by f, denoted by gof, that is $(gof)(x) = g(f(x))$
- **Vertical Line Test**: Vertical line test is used to determine whether a graph of a curve is a function or not. If any curve cuts a vertical line at more than one point then the curve is not a function.

Examples

1. Identify whether the following are functions or not.

   a. $R_1 = \{(1,-2), (3,4), (5,5), (-1,-2)\}$

   b. $R_2 = \{(x, y): x \text{ is the square of } y\}$

Solution:

R_1 is a function, because every first element is paired with exactly one element of the second element.

R_2 is not a function, because the ordered pair $\{(4,2),(4,-2)\}$ can't satisfy the definition of function.

2. If $f(x) = 3x - 5$ and $g(x) = 4x - 3$, then find $f(g(x))$

Solution: $f(x) = 3x - 5$

$$f\big(g(x)\big) = 3\big(g(x)\big) - 5 = 3(4x - 3) - 5 = 12x - 9 - 5 = 12x - 14$$

Exercise

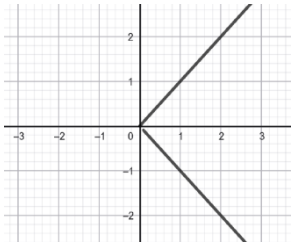
1. Identify whether the following are functions or not
   a. $R = \{(x, y): x + y^2 = 1\}$
   b. $R = \{(x, y): y < x\}$
   c. $R = \{(x, y): x = |y|\}$
   d. $R = \{(x, y): y = |x|\}$
2. If $f = \{(2,1),(3,-1),(-2,3),(0,7),(-1,-1)\}$ and $g = \{(7,-1),(1,0),(4,-2),(3,5),(-1,-3)\}$, then find fog

Exam type questions

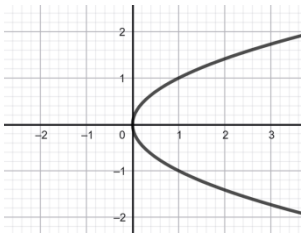
1. Which one of the following relations is a function?
   A. $R = \{(x, y): x = |y|, for\ x, y \in \mathbb{R}\}$
   B. $R = \{(x, y): x^2 + y^2 = 9, for\ x, y \in \mathbb{R}\}$
   C. $R = \{(x, y): y = x^{-0.5}, for\ x, y \in \mathbb{R}\}$
   D. $R = \{(x, y): x = 2, for\ x, y \in \mathbb{R}\}$
2. Let $f(x) = 4x + 1$ and $g(x) = 3x + b$. What is the value of b, so that $f\big(g(x)\big) = g\big(f(x)\big)$?
   A. $\frac{2}{5}$
   B. $\frac{1}{3}$
   C. $\frac{4}{7}$
   D. $\frac{2}{3}$

3. Which one of the following is the graph of a function?

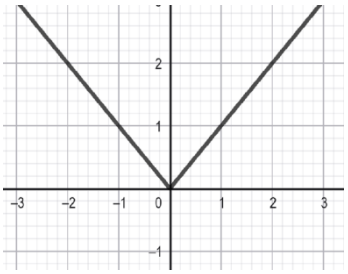
A.



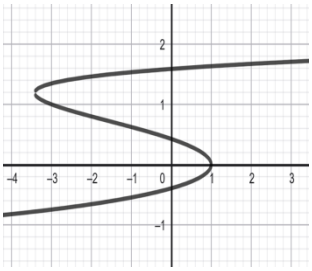
C.



B.



D.



Lesson 18 (day 18)

Types of functions (1-1, onto and 1-1 correspondence)

☞ By the end of this lesson students would be able to:

- Identify types of functions

Activity

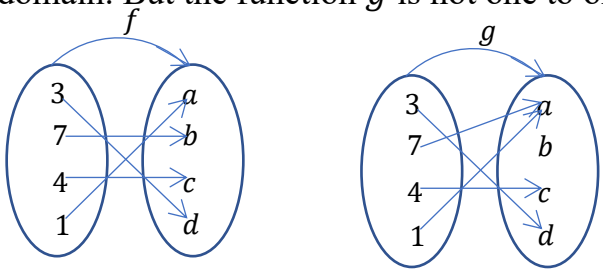
1. Let $R = \{(x, y): y \ is \ the \ father \ of \ x\}$ be a relation.

a. Is R a function? b. Is R one to one function?

Key concepts:

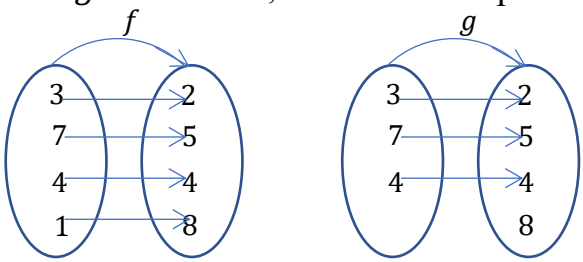
- **One-to-one function (Injective function):** If each element in the domain of a function has a distinct image in the co-domain, then the function is said to be one - to- one function.

Example: the function f below is one to one, since each element in the domain of f has a distinct image in the co-domain. But the function g is not one to one, since 7 and 1 has the same image a.



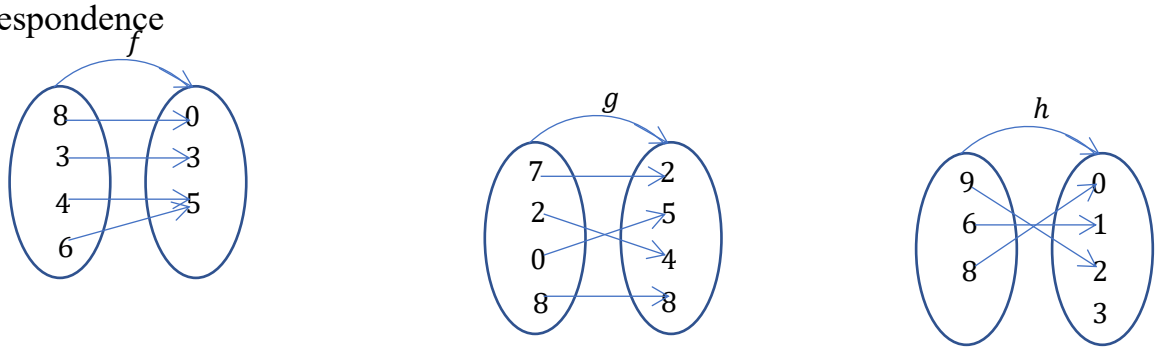
- A function $f: A \to B$ is called one to one if and only if for all $x, y \in A$, $f(x) = f(y)$ implies $x = y$.
- To test whether a given graph of a function is one to one or not, use horizontal line test. If a horizontal line crosses the graph at most once, then it is one to one otherwise it is not one to one function.
- **Onto function (Surjective Function):** A function is called an onto function if each element in the co-domain has at least one pre-image in the domain
- A function $f: A \to B$ is called on to if and only if the range of f is B.

Example: the function f below is on to, since every element in the codomain has pre-image. But, the function g is not on to, since 8 has no pre-image.



- **One-to-one correspondence (Bijective function):** A function $f: R \rightarrow R$ is said to be a one-to-one correspondence if f is both one-to-one and onto.

Example: identify whether the following functions are one to one, on to, one to one correspondence

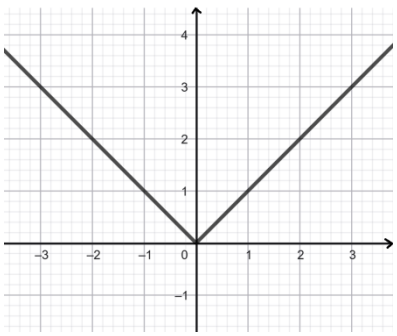


Solution:

- f is on to function but it is not one to one.
- g is both one to one and on to function, hence it is one to one correspondence.
- h is one to one function, but it is not on to.

Example: let $f: \mathbb{R} \rightarrow [0,\infty), f(x) = |x|$, is f one to one? , On to? , One to one correspondence?

Solution: sketch the graph of the function in the given domain



- The range of f is $[0,\infty)$, hence f is on to function
- A horizontal line crosses the graph more than once , hence, f is not on to function
- f is not one to one correspondence

Exam type questions

1. Which one of the following functions is one to one correspondence?

A. $f: \mathbb{R} \to \mathbb{R}, f(x) = tanx$

B. $f: \mathbb{R} \to \mathbb{R}, f(x) = 2^x$

C. $f: [0,\infty) \to [0,\infty), f(x) = x^2$

D. $f: [0,\infty) \to [0,\infty), f(x) = x + 5$

2. Which one of the following is one to one correspondence function from $A = [0,1]$ to $[1,2]$

A. $f(x) = x$

B. $f(x) = \frac{1}{3}x + 1$

C. $f(x) = 2x + 1$

D. $f(x) = x^2 + 1$

3. Which of the following functions is one to one correspondence?

A. $f: [0,\infty) \to \mathbb{R}, f(x) = |x|$

B. $f: \mathbb{R} \to [0,\infty), f(x) = x^2$

C. $f: \mathbb{R} \to [0,\infty), f(x) = 3^x$

D. $f: (0,\infty) \to \mathbb{R}, f(x) = \log_2 x$

Lesson 19 (day 19)

Inverse of function and their graph

☞ By the end of this lesson students would be able to:

- Find the inverse of a function

Key concepts:

- Given a function f, if it is one to one function, then we say that f is invertible and we denote its inverse by f^{-1}
- Given a function $f(x)$, if f is invertible then to find its inverse.
  - ✓ interchange x and y in the expression $y = f(x)$
  3. solve y in terms of x, (if possible);
  4. write f^{-1} as $y = f^{-1}(x)$
- $\mathrm{I}(x) = x$ is called identity function
- If $f(x)\ and\ g(x)$ are inverse to each other, then $f\big(g(x)\big) = g\big(f(x)\big) = x = \mathrm{I}(x)$
- The graph of $f^{-1}(x)$ is obtained from the graph of $f(x)$ by reflecting it along the line $y = x$.

Examples

1. Find the inverse of the following functions

a. $g(x) = 4x - 1$

b. $h(x) = \frac{2x-1}{x+1}$

Solution:

a. $g(x) = 4x - 1$

$y = 4x - 1$

$x = 4y - 1$

$4y = x + 1$

$y = \frac{x+1}{4}$

$g^{-1} = \frac{x+1}{4}$

b. $h(x) = \frac{2x-1}{x+1}$

$y = \frac{2x-1}{x+1}$

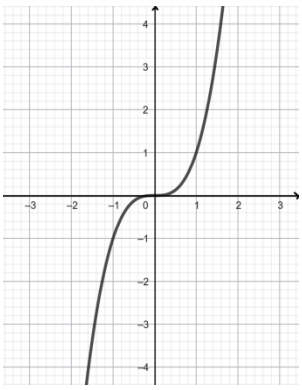
$x = \frac{2y-1}{y+1}$

$xy + x = 2y - 1$

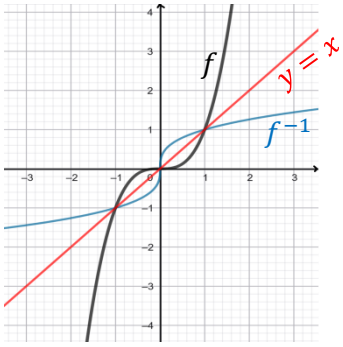
$y(x - 2) = -x - 1$

$h^{-1} = \frac{x+1}{2-x}$

2. Sketch the graph of f^{-1}, the graph of f is given below:



Solution: reflect the graph of f along the line $y = x$



Exam type questions

1. If $f(x) = ax - b$ and $f^{-1}(x+1) = \frac{1}{2}x + 2$, then what are the values of a *and* b respectively.

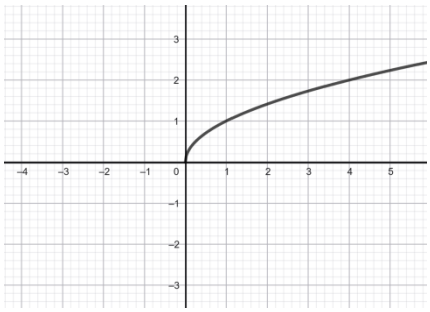
   A. 2 and 3 C. 5 and 7

   B. 1 and 3 D. 1 and 8

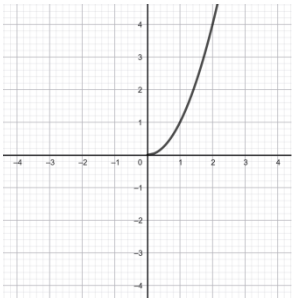
2. Given that $f(x) = \frac{1}{2}x - a$ and $g(x) = 3x + b$. If $g\big(g(2)\big) = 10$ and $f^{-1}(2) = 14$, then find the values of a and b respectively.

   A. 5 and -2 B. -5 and 3 C. -5 and -3 D. 5 and 2

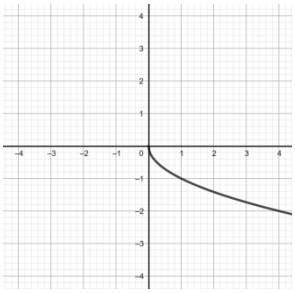
3. Which one of the following is the inverse of the graph below?



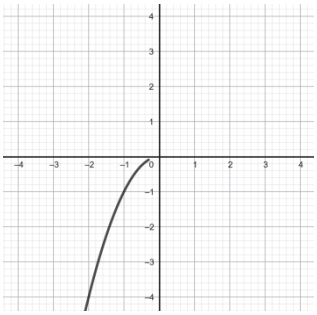
A.



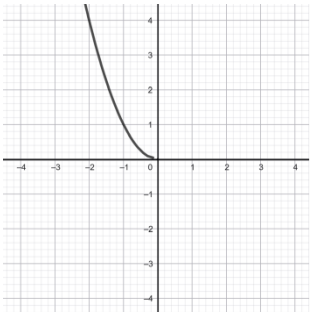
C.



B.



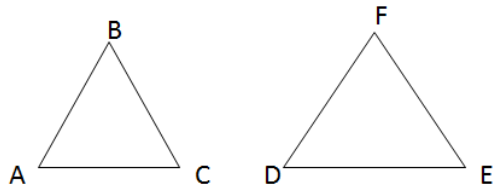
D.



Lesson 20(Day20)

Practice questions

1. In the figure shown blow $\angle BAC \equiv \angle DFE$, and $\angle ACB \equiv \angle FDE$ and $\overline{AC} \equiv \overline{DE}$ then which of the following is true?



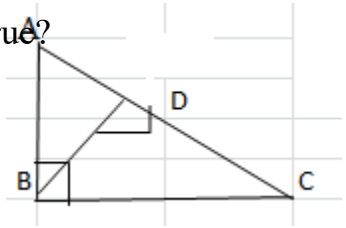
   A. $\Delta ABC \equiv \Delta DFE$

   B. $\Delta ABC \equiv \Delta DEF$

   C. $\Delta ABC \equiv \Delta EFD$

   D. $\Delta ABC \equiv \Delta FED$

2. Let ΔABC be aright angled triangle with the right angle at B. Let$\overline{BD}$ be perpendicular to $\overline{AC}$ (see figure blow) then which of the following is true?



   A. $\Delta ABC \sim \Delta ADB$

   B. $\Delta ABC \sim \Delta BDC$

   C. $\Delta DAB \sim \Delta DBC$

   D. $\Delta ADB \sim \Delta DBC$

3. Which one of the following pairs of function are inverse of each other in their respective domain?

   A. $f(x) = x^5$ and $g(x) = \frac{1}{x^5}$

   B. $f(x) = \sqrt{x+3}$ and $g(x) = x^2 - 9$

   C. $f(x) = \frac{x+1}{x-1}$ and $g(x) = \frac{x-1}{x+1}$

   D. $f(x) = x^3 - 1$ and $g(x) = \sqrt[3]{x+1}$

4. Which of the following is constant function?

   A, $f(x) = 2x$ C, $f(x) + 2x = 2 + 2x$

   B, $f(x) = 2 - 2x$ D, $f(x) = x + 1$

5. $f(x) = |x|, g(x) = x^2$. What is $f(g(x))$?

   A) $|x^2|$ B) x^2 C) x D) $|x|$

6. The graph of $f(g(x))$is obtained by:

   A) Applying g first, then f

   B) Applying f first, then g

   C) Reflecting across $y = x$

   D) multiplying f by g.

7. $f(x) = sin(x), domain = \mathbb{R}$ This function is:

A) One-to-one B) Onto (onto [-1,1]) C) Both D) Neither

8. $f(x) = \lfloor x \rfloor$ This function is:

A) One-to-one B) Onto (onto integers) C) Both D) Neither

9. $f(x) = tan(x), domain = (-\pi/2, \pi/2)$ This function is:

A) One-to-one only B) Onto only C) Both (bijective onto $\mathbb{R}$)

D) Neither

10. Let a function f be given as : $f(x) = \frac{1}{\sqrt{9 - x^2}}$ Domain is:

A) $(-3, 3)$ B) All real numbers C) $x > 0$ D) $(-\infty, \infty)$

11. Let $f(x) = \frac{x - 3}{x + 2)}$, the inverse of f is:

A) $f^{-1}(x) = \frac{2x+3}{1-x}$ B) $f^{-1}(x) = \frac{x+3}{x-2}$ C) $f^{-1}(x) = \frac{x-2}{x+3}$ D) None

12. The graph of an inverse function is always:

A) Reflected across the line $y = x$

B) Rotated 90° about the origin

C) Symmetric about the y-axis D) None

13. $f(x) = \sqrt{(x + 4}$. Which of the following is true?

A) $f^{-1}(x) = x^2 - 4$ B) $f^{-1}(x) = \sqrt{x - 4}$ C) $f^{-1}(x) = x^2 + 4$ D) None

14. Let $f(x) = 2x - 5$, $f\big(g(x)\big) = 2x^2 - 7x + 1$. Find $g(x)$

15. Let $(x) = 3x - 1$, $g(x) = x^2 - a$. Find a if $f\big(g(x)\big) = g(f(x))$

16. Let $(x) = ax + b$ and $f^{-1}(x + 3) = 2x + 7$. Find a and b.

Lesson 21 (Day 21)

Special types of functions

(power function, sgn function and greatest integer function).

☞ By the end of this lesson students would be able to:

- Work with a special types of functions

Key concepts:

1. **Power function:** is a function of the form $f(x) = ax^n$, where a and n are any real numbers and $a \neq 0$.
2. The function defined by $f(x) = sgnx = \begin{cases} 1, if\ x > 0 \\ 0, if\ x = 0 \\ -1, if\ x < 0 \end{cases}$ is called the **Signum function**.
3. The function $f(x) = \lfloor x \rfloor$ is the greatest integer that is less than or equal to x, it is called the **greatest integer** (**floor or step**) function.

Examples

1. Solve the following equations $sgn(x^2 + x + 4) = 1$

   We know that $sgnx = \begin{cases} 1, \text{if } x > 0 \\ 0, \text{if } x = 0 \\ -1, \text{if } x < 0 \end{cases}$

   So $sgn(x^2 + x + 4) = 1$ means $x^2 + x + 4 > 0$ which is always true. Therefore, the solution set is $\mathbb{R}$

2. What is the range of the function $f(x) = 6 - sgn\ x$?

   We have $6 - sgn\ x = 6 - \begin{cases} 1, \text{if } x > 0 \\ 0, \text{if } x = 0 \\ -1, \text{if } x < 0 \end{cases} = 6 - 1, 6 - 0, 6 - -1$. Which is 5, 6, 7

   Therefore, the range is $\{5, 6, 7\}$ and its domain is $\mathbb{R}$

Exercise

1. Solve the following equation $\lfloor x^2 - 1 \rfloor = 3$
2. What is the solution set of the equation $\lfloor x + 7 \rfloor = -8$?

3. Let $f(x) = \lfloor x \rfloor + sgn(2x)$, then $f(-0.5)$ is equal to:

Exam type questions

1. Which one of the following is true about signum, absolute value and greatest integer functions?

   A. $sgn(x) = \pm|x|$, for all $x \in \mathbb{R}$

   B. $sgn(x) \leq \lfloor x \rfloor$, for all $x \leq 0$

   C. $|x| = xsgn(x)$, for all $x \in \mathbb{R}$

   D. $sgn(x) \leq \lfloor x \rfloor$, for all $x \geq 0$

Lesson 22 (Day 22)

Theorem on polynomial functions and their graph

☞ By the end of this lesson students would be able to:

- Apply polynomial remainder theorem
- Apply polynomial factor theorem
- Identify the behaviors of the graph of polynomial function

Key concepts:

- If $f(x)$ and $d(x)$ are polynomials such that $d(x) \neq 0$, and the degree of $d(x)$ is less than or equal to the degree of $f(x)$, then there exists a unique polynomial $q(x)$ and $r(x)$ such that

$$f(x) = q(x).d(x) + r(x)$$

Dividend: $f(x)$; Quotient: $q(x)$; Divisor: $d(x)$; Remainder: $r(x)$

Where $r(x) = 0$ or the degree of $r(x)$ is less than the degree of $d(x)$.

- Let $f(x)$ be a polynomial of degree greater than or equal to 1 and let c be any real number. If $f(x)$ is divided by $(x - c)$, then the remainder is $f(c)$
- Let $f(x)$ be a polynomial of degree greater than or equal to one, and let c be any real number, then $x - c$ is a factor of $f(x)$ if and only if $f(c) = 0$.

Properties of the graph of polynomial function

- Degree n polynomial function has at most $n - 1$ turning points.
- Degree n polynomial function has at most n zeros.
- The $y -$ intercept of polynomial function is its constant term.

- If a is the zero of polynomial function with odd multiplicity, then the graph crosses $x-axis$ at $x=a$.
- If a is the zero of polynomial function with even multiplicity, then the graph touches $x-axis$ at $x=a$, but it doesn't cross $x-axis$ at this point.
- The graph of odd degree polynomial function with positive leading coefficient, moves far down from the left and move far up from the right
- The graph of odd degree polynomial function with negative leading coefficient, moves far up from the left and move far down from the right
- The graph of even degree polynomial function with positive leading coefficient, moves far up from the left and from the right
- The graph of even degree polynomial function with negative leading coefficient moves far down from the left and from the right.

Examples

1. When $f(x)=5x^3-kx^2+8x-1$ is divided by $d(x)=x-1$, the remainder is 10. Find the value of k.

Solution: $f(1)=10 \Rightarrow 5(1)^3-k(1)^2+8\times 1-1=10.$

$$\Rightarrow 5-k+8-1=10$$

$$\Rightarrow k=2$$

Exercise

1. If $x+3$ is a factor of $f(x)=x^4+2kx^3-x^2-5kx+6$, then find the value of k.
2. For the following of graph of polynomial function, determine

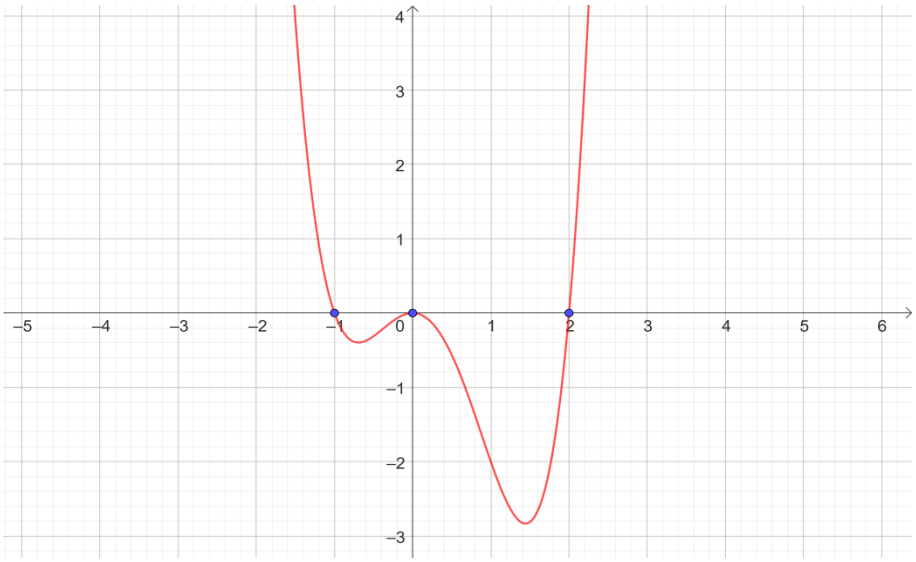
a. Number of turning points

b. Degree of the function

c. The sign of leading coefficient

d. The constant term of the function

e. The zeros of the function with their multiplicity (whether it is odd or even)



Exam type questions

1. Which one of the following a true statement?

A. If you divide $5x^3 + 4x^2 + x + 9$ by $x + 2$, then the remainder is -17

B. $x - 1$ is the factor of $x^n + 1$, for a natural number n

C. If $x^4 + 4x^2 + 3$ is divided by $x - 1$; the constant term of the quotient is 3

D. $x - 2$ is the factor of $x^4 + 2x^3 - 3x^2 - 4x + 4$

2. Which one of the following is **NOT** true about the graph of the polynomial function $f(x) = 2x^5 - 7x^3 + 2x - 1$?

A. It has at most 4 turning points and intersects the x-axis almost 5 times

B. As we move far from the origin it rises upward to the right and falls downward to the left

C. It is a smooth and continuous curve

D. It passes through the origin

3. Let $f(x) = x^3 + 5x^2 - x + 8$ and $g(x) = x^2 + 2$. If $f(x)$ is divided by $g(x)$, then

A. the remainder is $x + 5$

B. the quotient is $3x + 2$

C. the constant term of the quotient is -2

D. the degree of the remainder is 1

Lesson 23 (Day 23)

Graphs of Exponential Functions

☞ By the end of this lesson students would be able to:

- Draw the graph of exponential function

Key concepts:

- The exponential function f with base a is denoted by $f(x) = a^x$ where $a > 0$, $a \neq 1$ and x is any real number
- A distinct characteristic of an exponential function $f(x) = a^x$ is showing a rapid increase as x increases for $a > 1$ and showing a rapid decrease as x increases for $a < 1$
- The function $f(x) = 2^x$ is positive for all values of x.

☞ As x increases, the value of the function gets larger and larger.

☞ As x decreases, the value of the function get smaller and smaller, approaching zero.

- The function $f(x) = \left(\frac{1}{2}\right)^x$ is positive for all values of x.

  ☞ As x increases, the value of the function get smaller and smaller, approaching zero.

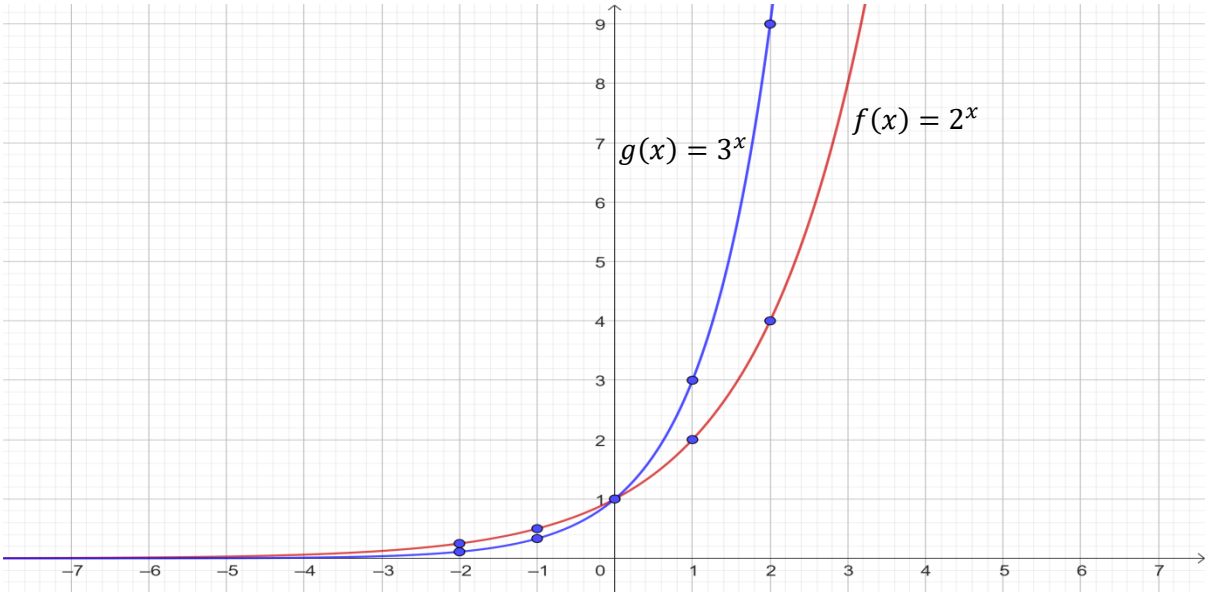
  ☞ As x decreases, the value of the function gets larger and larger.

- The natural exponential function is the exponential function $f(x) = e^x$ with base e and $2 < e < 3$

Examples

1. Sketch the graph of $f(x) = 2^x$ and $g(x) = 3^x$ on the same coordinate plane

Solution

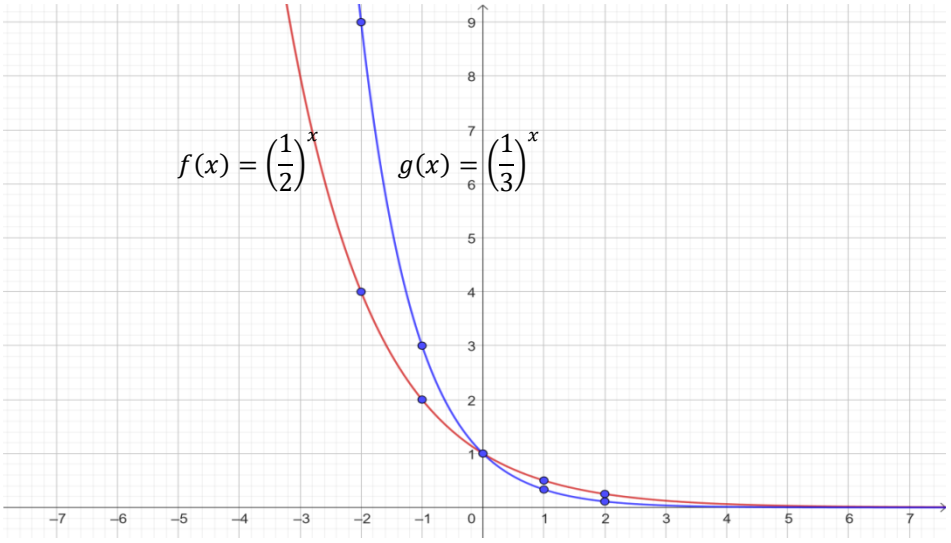


The basic characteristics of graphs of $f(x) = a^x, a > 1$

1. Domain: $\mathbb{R}$ = The set of all real numbers.
2. Range: $\mathbb{R}^+$ = The set of all positive real numbers or $\{y|y > 0, y \in \mathbb{R}\}$.
3. Y- intercept: the point (0,1) (or The graph crosses the y-axis at $y = 1$)
4. Has no x- intercept(*or* The graph never intersects the x-axis)
5. It is increasing on $\mathbb{R} = (-\infty, \infty)$
6. The values of y increase as the values of x Increase. Therefore, the function is increasing over its domain.
7. Since the graph approaches the line y =0 as the values of x get very small, y = 0 is the horizontal asymptote. When $x \to -\infty$ then f(x) $\to$ 0 and when $x \to \infty$ then $f(x) \to \infty$.

2. Sketch the graph of $f(x) = \left(\frac{1}{2}\right)^x$ and $g(x) = \left(\frac{1}{3}\right)^x$ on the same coordinate plane

Solution



The basic characteristics of the graph of $f(x) = a^x, 0 < a < 1$

1. Domain: $\mathbb{R}$ = The set of all real numbers.
2. Range: $\mathbb{R}^+$ = The set of all positive real numbers or $\{y|y > 0, y \in \mathbb{R}\}$.
3. Y- intercept: the point (0,1) (or The graph crosses the y-axis at $y = 1$)
4. Has no x- intercept(*or* The graph never intersects the x-axis)
5. It is decreasing function. The value of f decreases whenever the value of x increases.
6. Horizontal asymptote: The x-axis (the line $y = 0$) is a horizontal asymptote.

Exercise

1. Sketch the graph of $f(x) = 4^{-x}$

Exam type questions

1. Let $f(x) = \left(\frac{1}{3}\right)^x$ and $g(x) = \left(\frac{3}{4}\right)^x$. Which of the following is true about these functions?

   A. For $x \geq 0$, the graph of $f(x)$ lies above the graph of $g(x)$.

   B. For $x < 0$, the graph of $f(x)$ lies above the graph of $g(x)$.

   C. For $x < 0$, both the graphs of $f(x)$ and $g(x)$ lie below the line $y = 1$.

   D. For $x > 0$ both the graphs of $f(x)$ and $g(x)$ lie below the linen $y = 1$

Lesson 24 (Day 24)

Graphs of Logarithmic Functions

☞ By the end of this lesson students would be able to:

- Draw the graph of logarithmic function
- Draw the graph of exponential and logarithmic function on the same coordinate plane

Activity

1. If $f(x) = 3^x$, then find $f^{-1}(x)$.
2. If $f(x) = \log_2 x$, then find $f^{-1}(x)$.

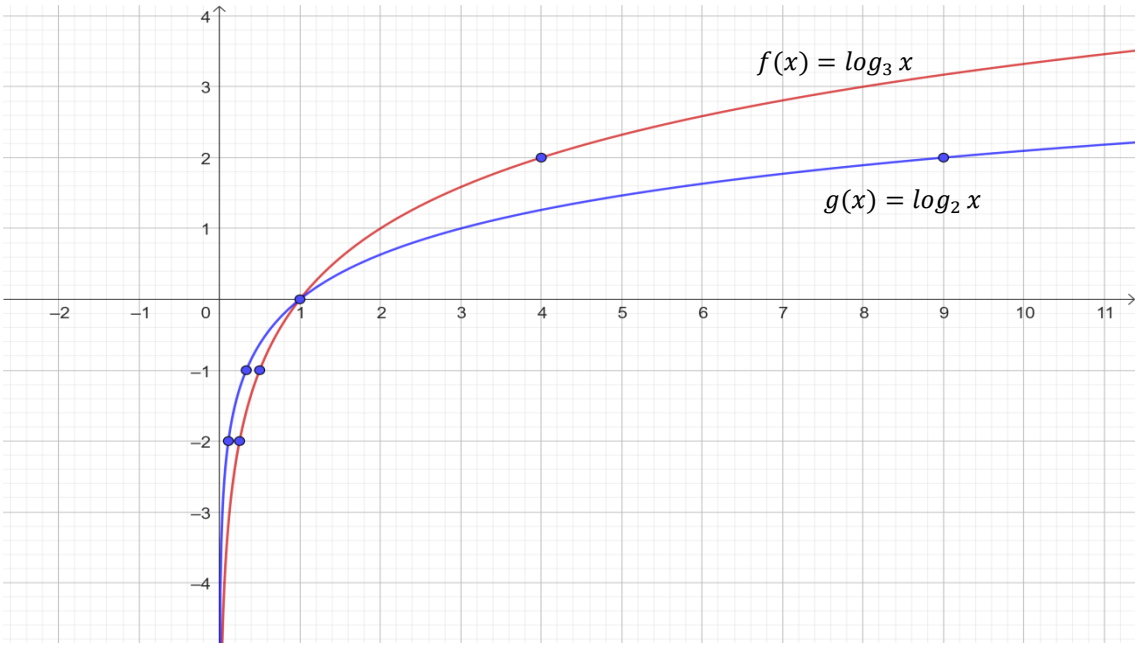
Key concepts:

- The inverse of the function $f(x) = a^x$ with $a > 0$ and $a \neq 1$ is $f^{-1}(x) = \log_a x$
- To sketch the graph of logarithmic functions, prepare a table by listing some possible values of y and calculate the corresponding values of x

Examples

1. Sketch the graph of $f(x) = \log_3 x$ and $g(x) = \log_2 x$ on the same coordinate plane

Solution

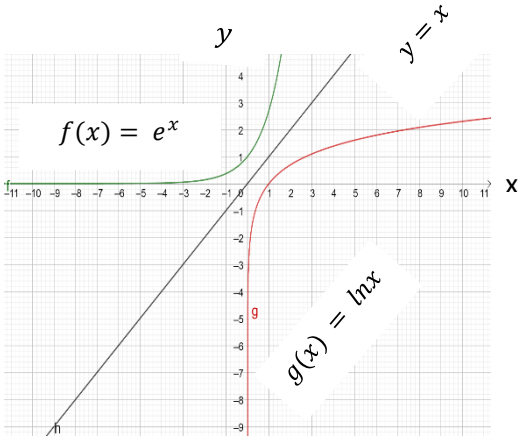


In general the function $g(x) = \log_a x$, a >1, have the following properties

1. The domain is all positive real numbers.
2. The range is all real numbers.
3. The graph contains the point (1,0).That is the x-intercept is (0,1) .
4. The graph does not intersect the y-axis.
5. The function is increasing. As x-increases y-increases.
6. As x gets closer and closer to 0 the value of the function gets closer and closer to the negative y-axis

2. Sketch the graph of $f(x) = e^x$ and $g(x) = lnx$ on the same coordinate plane

Solution:



Exercise

1. Sketch the graph of $f(x) = \log_{\frac{1}{3}} x$ and list the general characteristics of the graph of

   $f(x) = \log_a x$, for $0 < a < 1$

Exam type question

1. For $a>0$ and $a\neq 1$, which one of the following is true about $f(x)= a^x$, and $g(x)= \log_a x$?

A. They have the same range.

B. They have the same domain.

C. Their y- intercepts are the same.

D. The domain of f(x) is the same as the range of g(x).

Lesson 25 (Day 25)

Practice questions

1. Let $f(x) = x^{\frac{2}{3}}$. Range of f is:

   A) $[0, \infty)$ B) $(0, \infty)$ C) $(-\infty, 0]$ D) $\mathbb{R}$

2. Let $f(x) = x^{\frac{3}{2}}$. Domain of f is:

   A) $[0, \infty)$ B) $(-\infty, \infty)$ C) $(-\infty, 0]$ D) can't be determined

3. What is the simplified form of $f(x) = x \cdot sgn(x)$?

   A) |x| B) x C) −x D) 0

4. Let $f(x) = sgn(x^2 - 4)$. What if the Value of f at $x = 3$?

   A) 1 B) 0 C) −1 D) 5

5. Let $f(\boldsymbol{x}) = \boldsymbol{sgn(e^x)}$. Range of f is:

   A) {1} B) {−1, 1} C) {0, 1} D) {-1, 0, 1}

6. Let $f(x) = -x^4 + 3x^2 - 5$. Which of the following is the behavior of f as $x \to \infty$:

   A) $f(x) \to -\infty$ B) $f(x) \to \infty$ C) $f(x) \to 0$ D) can't be determined

7. Let $f(x) = (x - 2)^3(x + 1)$. which of the following is true?

   A) f crosses the x-axis at x=2

   B) f touches the x-axis at x=-1

   C) f touches the x-axis at x=2

   D) can't be determined

8. Let $f(x) = -x^3 + 2x$. behavior of f as $x \to \infty$ is:

   A) $f(x) \to -\infty$ B) $f(x) \to \infty$ C) $f(x) \to 0$ D) $f(x) \to -2$

9. If $x - 2$ is a factor of $f(x) = x^3 - 3kx^2 + 5kx + 8$, then what is the value of k.

   A) -4 B) 2 C) 8 D) -12

10. What is the horizontal asymptote of $f(x) = 2^x - 5$?

A) $y = -5$ B) $y = 0$ C) $y = 5$ D) $y = -4$

11. Let $f(x) = e^{(x+3)}$. Graph of f is obtained from $g(x) = e^x$ by shifting:

A) g left by 3 B) g right by 3 C) g up by 3 D) can't be determined

12. Which of the following is equal to the expression $\frac{(125)^{\frac{2}{3}}(81)^{\frac{-3}{4}}}{(27)^{-\frac{2}{3}}(625)^{\frac{3}{4}}}$?

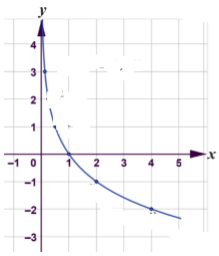
A. 15 B. $\frac{1}{15}$ C. $\frac{5}{3}$ D. $\frac{3}{5}$

13. The simplified form of $\frac{9^{\sqrt{3}} \times 81^{\sqrt{12}}}{27^{\sqrt{48}}}$ is _____?

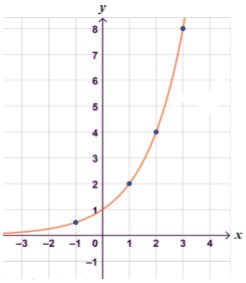
A. $3^{-2\sqrt{3}}$ B. $3^{\sqrt{3}}$ C. $3^{2\sqrt{3}}$ D. 3

14. Which one of the following graph represent $f(x) = \log_4 x$?

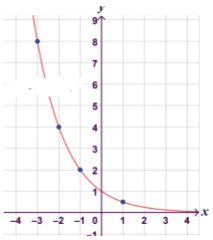
A.



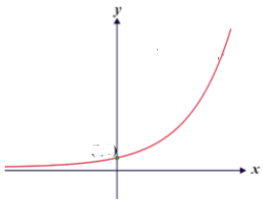
B.



C.



D.



15. Solve for x if $4^x = 2^{(2x-1)}$.

16. The simplified form of $\log_a 8 \times \log_{64} a^6$ is_____

Lesson 26 (Day 26)

Solving Logarithmic Equations

☞ By the end of this lesson students would be able to:

- Find solution to logarithmic equations

Key concepts:

- For any positive real numbers x, y, $a > 0$ and $a \neq 1$
  $\log_a x = \log_a y$, if and only if $x = y$.

We use the following procedures to solve logarithmic equations.

- State the universe.
- Collect the logarithmic term on one side of the equation.
- Write the equation in exponential form.
- Solve for the variable.

Examples:

1. Solve the following logarithmic equations

a. $\log_2(3x-1) = 5$
b. $\log_{\sqrt{2}} x = 6$
c. $\log_2(x^2 - 3x) = 4$
d. $\log(x^2 - 121) - \log(x+11) = 1$
e. $\log_x(x+6) = 2$

Solutions:

a. $\log_2(3x-1) = 5$, The universe of the equation $x > \frac{1}{3}$
Convert the logarithm into exponent
$\text{Log}_2(3x-1) = 5 \Leftrightarrow 3x - 1 = 2^5$
$3x - 1 = 32$
$3x = 32 + 1$
$3x = 33 \Rightarrow x = 11, \quad \text{s.s} = \{11\}$

b. $\log_{\sqrt{2}} x = 6$, The universe is $x > 0$
Convert the logarithm statement into exponential statement

$$x = \left(\sqrt{2}\right)^6$$
$$x = (2)^{\frac{6}{2}}$$

$$x = (2)^3$$
$$x = 8$$
$$\text{s.s} = \{8\}$$

c. $\log_2(x^2 - 3x) = 4$

The universe is $x < 0 \ \cup \ x > 3$

$\log_2(x^2 - 3x) = 4$ Convert to exponential statement

$x^2 - 3x = 2^4$

$x^2 - 3x - 16 = 0$

Using quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-(-3) \pm \sqrt{(-3)^2 - 4 \times 1 \times (-16)}}{2(1)}$$

$$x = \frac{3 \pm \sqrt{9 + 64}}{2}$$

$x = \frac{3 \pm \sqrt{73}}{2}$ $\Rightarrow x = \frac{3+\sqrt{73}}{2}$ and $x = \frac{3-\sqrt{73}}{2}$

$$s.s = \left\{\frac{3+\sqrt{73}}{2}, \frac{3-\sqrt{73}}{2}\right\}$$

d. $\log(x^2 - 121) - \log(x + 11) = 1$

the universe is $(11, \infty)$

$\log((x - 11)(x + 11)) - \log(x + 11) = 1$

$\log\left(\frac{(x-11)(x+11)}{(x+11)}\right) = 1$ Convert to exponential statement

$\frac{(x-11)(x+11)}{(x+11)} = 10^1$

$x - 11 = 10$

$x = 21 \ \in (11, \infty)$

$s.s = \{21\}$

e. $\log_x(x+6)=2$

The universe is $x>0\,, x\neq 1$

$\log_x(x+6)=2$ Convert to exponential statement

$x^2=x+6$

$x^2-x-6=0$

$\Rightarrow (x-3)(x+2)=0$

$\Rightarrow x=3\ and\ x=-2$

$\boldsymbol{s}.s=\{3\}$, -2 is not in the universe

Exercises

1. Solve the following logarithmic equations

A. $\log x-\log 3=\log 4-\log(x+4)$
B. $\log_2 2+\log_2(x+2)-\log_2(3x-5)=3$
C. $\log_2\left(1+\frac{1}{x}\right)=3$

Lesson 27 (Day 27)

The Six Trigonometric Ratios

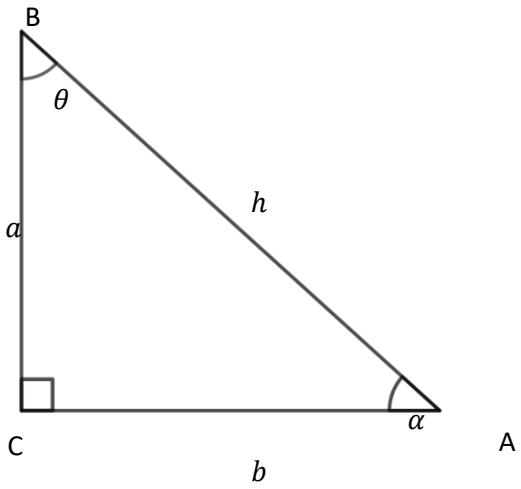
By the end of this lesson students would be able to:

- Determine Trigonometric ratios (values) of various angles

Key concepts:

Basic Trigonometric ratios

- For any right-angle triangle ΔABC, with an acute angle α and θ, the sine, cosine and tangent ratios will be defined as follows:



$$\sin\alpha = \frac{opposite}{hypoutnouse} = \frac{BC}{AB} = \frac{a}{h} \qquad \sin\theta = \frac{opposite}{hypoutnouse} = \frac{AC}{AB} = \frac{b}{h}$$

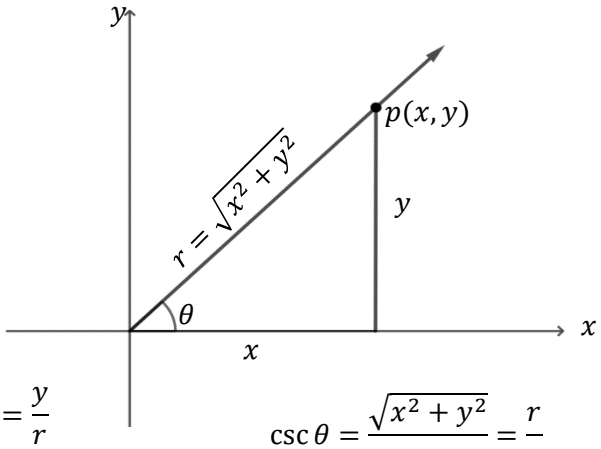
$$\cos\alpha = \frac{adjecent}{hypoutnouse} = \frac{AC}{AB} = \frac{b}{h} \qquad \cos\theta = \frac{adjecent}{hypoutnouse} = \frac{BC}{AB} = \frac{a}{h}$$

$$\tan\alpha = \frac{opposite}{adjecent} = \frac{BC}{AC} = \frac{a}{b} \qquad \tan\theta = \frac{opposite}{adjecent} = \frac{AC}{BC} = \frac{b}{a}$$

The reciprocals of basic trigonometric ratios

- The reciprocal of sine is cosecant that is $\csc\theta = \frac{1}{\sin\theta}$
- The reciprocal of cosine is secant that is $\sec\theta = \frac{1}{\cos\theta}$
- The reciprocal of tangent is cotangent that is $\cot\theta = \frac{1}{\tan\theta}$

The values of sine, cosine and tangent and their reciprocal ratios when the terminal side of the angle passes through a point $p(a, b)$ is calculated as follows:



$$\sin\theta = \frac{y}{\sqrt{a^2+b^2}} = \frac{y}{r} \qquad \csc\theta = \frac{\sqrt{x^2+y^2}}{y} = \frac{r}{y}$$

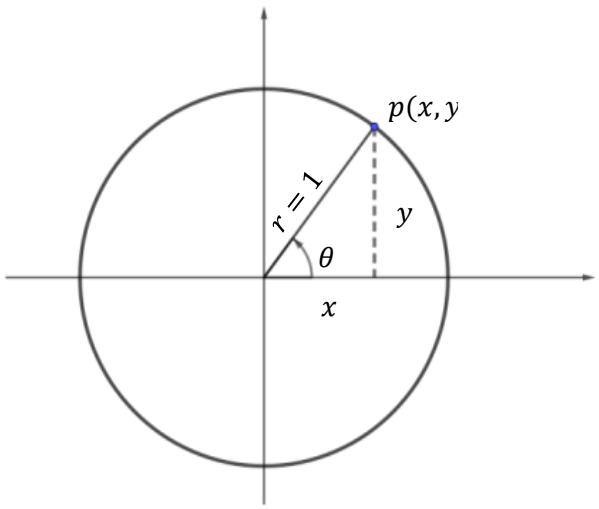
$$\cos\theta = \frac{x}{\sqrt{x^2+y^2}} = \frac{x}{r} \qquad \sec\theta = \frac{\sqrt{x^2+y^2}}{x} = \frac{r}{x}$$

$$\tan\theta = \frac{y}{x} \qquad \cot\theta = \frac{x}{y}$$

Quadrantal angles are angles whose the terminal sides lies on the x or y axis.

To find trigonometric values of quadrantal angles use a unit circle (a circle with a radius of 1) and centered at the origin.

Consider a point $p(x, y)$ on a unit c



$$\sin\theta = \frac{y}{r} = \frac{y}{1} = y$$

$$\cos\theta = \frac{x}{r} = \frac{x}{1} = x$$

$$\tan\theta = \frac{y}{x}$$

$$\csc\theta = \frac{r}{y} = \frac{1}{y}$$

$$\sec\theta = \frac{r}{x} = \frac{1}{x}$$

$$\cot\theta = \frac{x}{y}$$

Examples

1. Evaluate the sine, cosine, tangent, secant, cosecant and cotangent of angle θ if θ is in standard position and its terminal side contains the point $p\left(\frac{-\sqrt{3}}{2}, \frac{1}{2}\right)$.

Solution:

$$\mathrm{r} = \sqrt{x^2 + y^2} = \sqrt{\left(\frac{-\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = \sqrt{\frac{4}{4}} = 1$$

$$\sin\theta = \frac{b}{r} = \frac{\frac{1}{2}}{1} = \frac{1}{2}$$

$$\csc\theta = \frac{1}{\sin\theta} = \frac{1}{\frac{1}{2}} = 2$$

$$\cos\theta = \frac{a}{\sqrt{a^2+b^2}} = \frac{\frac{-\sqrt{3}}{2}}{1} = \frac{-\sqrt{3}}{2}$$

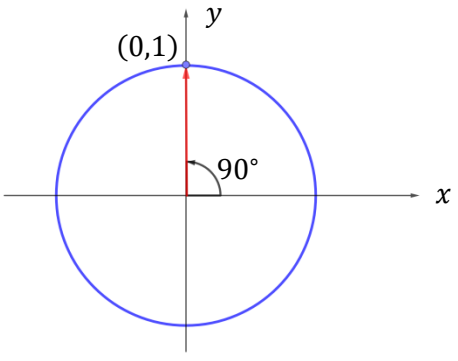
$$\sec\theta = \frac{1}{\cos\theta} = \frac{1}{\frac{-\sqrt{3}}{2}} = -\frac{2}{\sqrt{3}} = -\frac{2\sqrt{3}}{3}$$

$$\tan\theta = \frac{b}{a} = \frac{\frac{1}{2}}{\frac{-\sqrt{3}}{2}} = -\frac{1}{\sqrt{3}} = -\frac{\sqrt{3}}{3}$$

$$\cot\theta = \frac{a}{b} = \frac{\frac{-\sqrt{3}}{2}}{\frac{1}{2}} = -\sqrt{3}$$

2. using a unit circle find the six trigonometric values of $90°$

Solution: the terminal side of $90°$ intersects the unit circle at$(0,1)$



$$\sin 90° = \frac{y}{r} = \frac{1}{1} = 1$$

$$\cos 90° = \frac{x}{r} = \frac{0}{1} = 0$$

$$\tan 90° = \frac{y}{x} = \frac{1}{0} = \text{undefined}$$

$$\csc 90° = \frac{r}{y} = \frac{1}{1} = 1$$

$$\sec 90° = \frac{r}{x} = \frac{1}{0} = \text{undefined}$$

$$\cot 90° = \frac{x}{y} = \frac{0}{1} = 0$$

Exercise

1. Evaluate the sine, cosine, tangent, secant, cosecant and cotangent of angle θ if θ is in standard position and its terminal side contains the point $p(1,-1)$

2. Find the six trigonometric values for the angle -180^0

Exam type questions

1. What is the value of $\cot 270^0 + 2\cos 90^0 + 4\sec 180^0$
   A. -2 B. 8 C. 7 D. -4
2. Which one of the following is true about a second quadrant angle θ in standard position whose terminal side lies on the line $2x + y = 0$?
   A. $\cos\theta = \frac{2}{\sqrt{5}}$ B. $\sin\theta = \frac{1}{\sqrt{5}}$ C. $\cos\theta = \frac{1}{\sqrt{5}}$ D. $\sin\theta = \frac{2}{\sqrt{5}}$
3. If $\sin\theta = \frac{2}{3}$ for an acute angle θ, then which one of the following is correct?
   A. $\cot\theta = \frac{4\sqrt{5}}{5}$
   B. $\tan\theta = 2$
   C. $\csc\theta = \frac{6\sqrt{5}}{5}$
   D. $\sec\theta = \frac{3\sqrt{5}}{5}$

Lesson 28 (Day 28)

Co-terminal angles, reference angle and complementary angles relations

By the end of this lesson students would be able to:

- Identify co-terminal angles
- Determine reference angle for a given angle
- Identify trigonometric relation for complementary angles

Key concepts:

Co-terminal angles are angles that have the same terminal sides.

- Co-terminal angles for an angle θ is $\theta \pm n \times 360^0$ or $\theta = \theta \pm n \times 2\pi$, where n is any integer

The reference angle$(\boldsymbol{\theta_R})$ of an angle θ is an acute angle formed by the terminal side of the angle and x-axis.

For complementary angles θ and α, the following holds true:

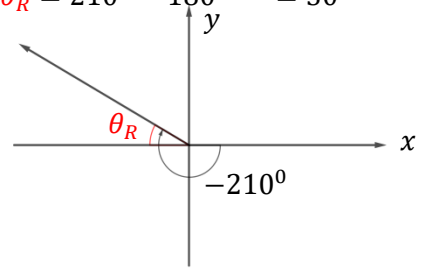
- $\sin\theta = \cos\alpha$ $\quad \tan\theta = \cot\alpha$ $\quad \sec\theta = \csc\alpha$

Examples

1. Find the reference angle θ_R for $\theta = -210^0$

Solution:

$\theta_R = 210^0 - 180^0 \quad = 30^0$



2. Find two co-terminal angles for 30^0

Solution:

- $30^0 + 360^0 = 390^0$
- $30^0 - 360^0 = -330^0$

3. If $\sin(2\theta) = \cos\theta$, then find the value of θ.

Solution: sine and cosine are equal for complementary angles,

$$2\theta + \theta = 90^\circ$$
$$\Rightarrow 3\theta = 90^\circ$$
$$\Rightarrow \theta = 30^\circ$$

Exercise

1. Find one positive and one negative co-terminal angle for -60°
2. If $sin\ 2A\ =\ cos\ (A\ -\ 30°)$, where $2A$ is an acute angle, find the value of A.
3. Find the reference angle for the angle 520°

Exam type questions

1. Which of the following is the measure of an angle that is co-terminal with an angle of -125^0?

   A. $\frac{25}{30}\pi$ radian B. -55° C.235° D. $\frac{-47}{36}\pi$ radian

2. Which one of the following is **NOT true** for$0 \le \theta \le \frac{\pi}{2}$?

   A. $\sec\theta = \csc\left(\frac{\pi}{2} - \theta\right)$

   B. $\csc\theta = \sin\left(\frac{\pi}{2} - \theta\right)$

   C. $\tan\theta = \cot\left(\frac{\pi}{2} - \theta\right)$

   D. $\cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$

Lesson 29

Trigonometric Identities and Equations

By the end of this lesson students would be able to:

- Solve trigonometric equations
- Apply half and double angle formula

Key concepts:

The trigonometric identities

- $sin^2\theta + cos^2\theta = 1$
- $1 + tan^2\theta = sec^2\theta$
- $1 + cot^2\theta = csc^2\theta$
- $\tan\theta = \frac{\sin\theta}{\cos\theta}$
- $\cot\theta = \frac{\cos\theta}{\sin\theta}$

Double angle identities

- $\sin 2\theta = 2\sin\theta\cos\theta$
- $\cos 2\theta = cos^2\theta - sin^2\theta$
  $= 1 - sin^2\theta - sin^2\theta$
  $= 1 - 2sin^2\theta$
- $\cos 2\theta = cos^2\theta - sin^2\theta$
  $= cos^2\theta - (1 - cos^2\theta)$
  $= 2cos^2\theta - 1$
- $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$

Half angle identities

- $\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$
- $\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$
- $\tan\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}$

Examples:

1. What is the value of $\sin 15^{\circ}$

Solution:

$$\sin 15^{\circ} = \sin\left(\frac{30^{\circ}}{2}\right) = \sqrt{\frac{1-\cos 30}{2}} = \sqrt{\frac{1-\frac{\sqrt{3}}{2}}{2}} = \sqrt{\frac{2-\sqrt{3}}{2}}$$

2. Suppose θ is first quadrant angle and $\tan\theta = \frac{2}{3}$, then find $\sec\theta$

Solution:

Using the trigonometric identity $1 + tan^2\theta = sec^2\theta$

$1 + \left(\frac{2}{3}\right)^2 = sec^2\theta \Rightarrow sec^2\theta = \frac{13}{9} \Rightarrow \sec\theta = \pm\frac{\sqrt{13}}{3}$, since θ is in first quadrant

$\sec\theta = \frac{\sqrt{13}}{3}$

3. Suppose θ is second quadrant angle and $\sin\theta = \frac{3}{4}$, then find $\sin 2\theta$.

Solution:

$\sin 2\theta = 2\sin\theta\cos\theta$, since $\sin\theta$ is given, we need to find $\cos\theta$

$$sin^2\theta + cos^2\theta = 1 \Rightarrow \left(\frac{3}{4}\right)^2 + cos^2\theta = 1 \Rightarrow cos^2\theta = 1 - \frac{9}{16} \Rightarrow \cos\theta$$

$$= \pm\sqrt{\frac{7}{16}} = \pm\frac{\sqrt{7}}{4}$$

Since θ is second quadrant angle$\cos\theta = -\frac{\sqrt{7}}{4}$

Therefore, $\sin 2\theta = 2\sin\theta\cos\theta = 2\left(\frac{3}{4}\right)\left(-\frac{\sqrt{7}}{4}\right) = -\frac{3\sqrt{7}}{8}$

4. Solve the equation $2\sin\theta = 1, for\ 0^{\circ} \leq \theta \leq 360^{\circ}$

Solution:

$2\sin\theta = 1 \Rightarrow \sin\theta = \frac{1}{2}$, sine is positive in first and second quadrant

Hence, $\theta = 30^{\circ}$, the angle in first quadrant

And $\theta = 150^{\circ}$, in the second quadrant whose reference angle is $\theta = 30^{\circ}$

Therefore, the general solution for the given equation is $\{30^\circ \pm 360^\circ n, 30^\circ \pm 360^\circ n, where\ n \in \mathbb{Z}\}$

The solution for the equation on the given interval is $\{30^\circ, 150^\circ\}$

Exercise

1. Solve the equation $2\cos\theta = -\sqrt{3}, for\ 0^\circ \leq \theta \leq 360^\circ$
2. Suppose θ is second quadrant angle and $\cos\theta = -\frac{1}{2}$, then find $\cos 2\theta$.
3. What is the value of $\cos 15^\circ$

Exam type questions

1. Suppose θ is third quadrant angle and $\sin\theta = -\frac{1}{4}$, then find $\cos 2\theta$.

   A. $\frac{7}{8}$ B. $\frac{7}{16}$ C. 1 D. $\frac{14}{5}$

2. For a non-quadrantal angle θ , what is the simplified form of θ $cot^2\theta - csc^2\theta$?

   A. 1 B. $2cot^2\theta$ C. -1 D. $-2sec^2$

Lesson 30(day 30)

Practice questions

1. If $\log_2 x = \frac{1}{3}\log_2 8 - 2\log_2 \sqrt{3}$ then x is equal to _____?

   A. $\frac{\sqrt{3}}{2}$ B. $\frac{2}{3}$ C. $\frac{3}{2}$ D. $\frac{\sqrt{3}}{3}$

2. $2^{(2+\log_2 3)}$ is equal to____?

   A. 7 B. 6 C. 8 D. 12

3. What is solution of $\log_3(x+1) - \log_3(x-1) = 2$

   A. 10 B. $\frac{2}{3}$ C. 8 D. $\frac{5}{4}$

4. What is solution of $\log_5(x) + \log_5(x-4) = 2$

   A) 5 B) 6 C) 7 D) 3

5. What is solution of $\ln(x^2 + 1) = 2$

   A) $\pm\sqrt{e^2 - 1}$ B) $\pm\sqrt{(e - 1)}$ C) $\pm\sqrt{(2e - 1)}$ D) $\sqrt{x^2 + 1}$

6. What is solution of $\log_3(x + 1) - \log_3(x - 2) = 1$

   A) 5 B) 4 C) 3 D) 2

7. Let $tan\theta = \frac{\sqrt{5}}{2}$ and θ be an acute angle. What is the value of $sin\theta$?

   A. $\frac{\sqrt{5}}{3}$ B. $\frac{\sqrt{5}}{9}$ C. $\frac{2}{3}$ D. $\frac{3}{2}$

8. If $\cos\theta = \frac{2}{3}$ for an acute angle θ, then which one of the following is correct?

   A. $\sin\theta = \frac{\sqrt{5}}{6}$ B. $\tan\theta = \frac{\sqrt{5}}{2}$ C. $\sin(90° - \theta) = \frac{2}{3}$ D. $\sin\theta < \cos\theta$

9. Two poles of height 8m and 17m stands on a plane ground and the distance between them is 12m. What is the distance between their tops?

   A. 15m B. 12m C. 14m D. 20m

10. What is the solution set of $sin(2x) = \frac{\sqrt{3}}{2}$?

A) $x = \frac{\pi}{6} + n\pi, \frac{\pi}{3} + n\pi$ C) $x = \frac{\pi}{3} + n\pi, \frac{2\pi}{3} + n\pi$

B) $x = \frac{\pi}{12} + n\pi, \frac{5\pi}{12} + n\pi$ D) None

11. What is the solution set of $cos^2(x) - sin^2(x) = 0$?

A) $x = \frac{\pi}{4} + n\pi$ B) $x = \frac{\pi}{2} + n\pi$

C) $x = \frac{3\pi}{4} + n\pi$ D) None

12. What is the solution set of $2cos^2(x) - 3sin(x) = 0$?

A) $x = \frac{\pi}{6}, \frac{5\pi}{6}$ B) $x = \frac{\pi}{3}, \frac{2\pi}{3}$ C) $x = \frac{\pi}{4}, \frac{3\pi}{4}$ D) None

13. Simplify the following trigonometric expressions.

a. $\sin^2(x)\cos^2(x)$ b. $\tan^2(x) - \sec^2(x)$

14. What is the value of $-4\tan 270^0 + 2\sec 90^0 + 6\cos 180^0$

15. Find two co-terminal angles for 65^0

16. Find one positive and one negative co-terminal angle for 30°

17. Suppose θ is fourth quadrant angle and $\sin\theta = -\frac{1}{4}$, then find $\tan 2\theta$